Layer 1
Data Acquisition

Dispatcher
- IAM role
- Managed policy attachment
- Lambda function
- IoT topic rule
- Lambda invoke permission

Shared IoT policy
once per twin
- IoT policy
- IoT policy version

IoT device identity
per device (D)
- IoT Thing
- IoT certificate
- Certificate attachment
- Policy attachment
- Local auth files

Initial values
per device (V)
- Init value
- MQTT publish

Layer 2
Data Storage

Hot storage
- DynamoDB table
- DynamoDB backup

Hot-to-Cold Mover
- IAM role
- Managed policy attachment
- Inline policy
- Lambda function
- EventBridge rule
- EventBridge target
- Lambda invoke permission

Cold storage
- S3 bucket

Cold-to-Archive Mover
- IAM role
- Managed policy attachment
- Inline policy
- Lambda function
- EventBridge rule
- EventBridge target
- Lambda invoke permission

Archive storage
- S3 bucket

Hot Reader
- IAM role
- Managed policy attachment
- Inline policy
- Lambda function
- Lambda invoke permission

Layer 3
Data Processing

Device processor
per device (D)
- IAM role
- Managed policy attachment
- Lambda function

Persister
- IAM role
- Managed policy attachment
- Inline policy
- Lambda function

Event Checker
- IAM role
- Managed policy attachment
- Inline policy
- Lambda function

Event Feedback
- IAM role
- Managed policy attachment
- Inline policy – if feedback (F)
- Lambda function

Lambda Chain
- IAM role
- Managed policy attachment
- Step Functions state machine

Event action
per internal action (A)
- Event action
- IAM role
- Managed policy attachment
- Lambda function

Layer 4
Twin Management

TwinMaker workspace
- S3 bucket
- S3 CORS config
- S3 bucket policy
- IAM role
- Inline policy
- TwinMaker workspace
- TwinMaker scene cleanup (cleanup)
- TwinMaker entities (cleanup)
- TwinMaker component type (cleanup)

Device component type
per distinct type (T ≤ D)
- TwinMaker component type
- TwinMaker components (cleanup)

Entity hierarchy
per entity / component (H, C)
- TwinMaker hierarchy
- TwinMaker entities
- TwinMaker components

Layer 5
Visualization

Grafana workspace
optional (G): DEPLOY_MANAGED_GRAFANA
- IAM role
- Trust policy
- Inline policy
- Managed Grafana workspace

Resource family — Access (IAM roles, policies) · Compute (Lambda, Step Functions) · IoT Core (rules, identities, messages) · Data stores (DynamoDB, S3) · Scheduling (EventBridge) · Twin and visualization (TwinMaker, Grafana) · Configuration and local files

Hollow marker: lifecycle artifact (invoke permission, backup, cleanup, publish). Dashed card: optional.

Arrow A → B: a resource of B is created after one of A.